

AURA: UNIFIED MULTIMODAL FRAMEWORK FOR CONVERSATIONAL MUSIC EDITING

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ABSTRACT

Instruction-guided music editors typically process each request independently, limiting their ability to support workflows in which users progressively refine a track. We introduce AURA, a unified multimodal framework for conversational music editing. AURA uses a multimodal large language model to interpret the complete dialogue history, an optional image, and reference audio, distilling the editing intent into compact concept tokens. A concept-to-audio module injects these tokens and frame-aligned reference features into a frozen MusicGen backbone, enabling precise edits while preserving unaffected content. AURA optimizes only 91M parameters while retaining 1.9B frozen backbone parameters. Experiments on Slakh2100 and MoisesDB demonstrate substantial improvements in edit correctness and content preservation over existing instruction-guided methods, including a 4–5× reduction in FAD for out-of-domain addition and removal. <https://openrb-lab.github.io/AURA-demo/>

Index Terms— Music editing, multimodal large language models, controllable music generation, audio conditioning

1. INTRODUCTION

Recent advances in text-to-music generation have substantially expanded the possibilities for AI-assisted composition and production. Recent works [1, 2, 3, 4, 5, 6, 7, 8, 9] generate realistic music from textual descriptions using autoregressive or diffusion-based models. This progress has motivated a shift from generation toward controllable music editing, where users modify an existing track through natural-language or visual instructions while preserving its unaffected content.

Recent methods, including AUDIT [10], InstructME [11], M²UGen [12], Instruct-MusicGen [13], and LeVo 2 [14], support instruction-guided editing and stem-level operations such as adding, removing, or extracting musical sources. However, these systems generally process each request independently and therefore cannot resolve editing instructions that depend on earlier dialogue turns. This limitation is particularly restrictive in realistic production workflows, where users progressively refine a track and expect each edit to remain consistent with the evolving conversation.

To address this limitation, we introduce AURA, a unified multimodal framework for conversational music editing.

AURA uses a multimodal large language model to jointly interpret the complete dialogue history, an optional image, and reference audio. It distills the editing intent into concept tokens, which are projected into the conditioning space of a frozen MusicGen backbone. A concept-to-audio (C2A) module is proposed to combine these semantic representations with reference-music features, enabling the decoder to apply the requested edit while preserving musicality prior. Only the LoRA adapters [15], projectors, and C2A modules are trainable, retaining the pretrained generative prior of the backbone.

In summary, our main contributions are:

- We propose concept-guided music decoding, which represents multimodal editing intent using compact concept tokens and injects them into a pretrained music decoder through a C2A module. The module combines semantic edit control with frame-aligned reference conditioning to preserve unedited content.
- We introduce AURA, a unified multimodal framework that jointly reasons over dialogue history, images, and reference audio for coherent multi-turn music editing. AURA substantially improves edit correctness and content preservation across in-domain and out-of-domain.

We train AURA on agent-generated editing conversations constructed from the Slakh2100 training set [16] and evaluate it on Slakh2100-test as an in-domain benchmark and MoisesDB [17] as an out-of-domain benchmark. Experiments on single- and multi-turn editing demonstrate strong generalization and substantially better content preservation than existing instruction-guided approaches.

2. METHOD

2.1. Overall framework

Given an optional image I , a multi-turn conversation history $\mathcal{T}$, and a reference audio signal X_{ref} , AURA generates an edited audio signal $\hat{X}_{\text{tgt}}$:

$$\hat{X}_{\text{tgt}} = g(I, \mathcal{T}, X_{\text{ref}}). \quad (1)$$

AURA separates multimodal intent understanding from audio generation. A multimodal large language model (MLLM) infers the desired edit from the complete conversation and input

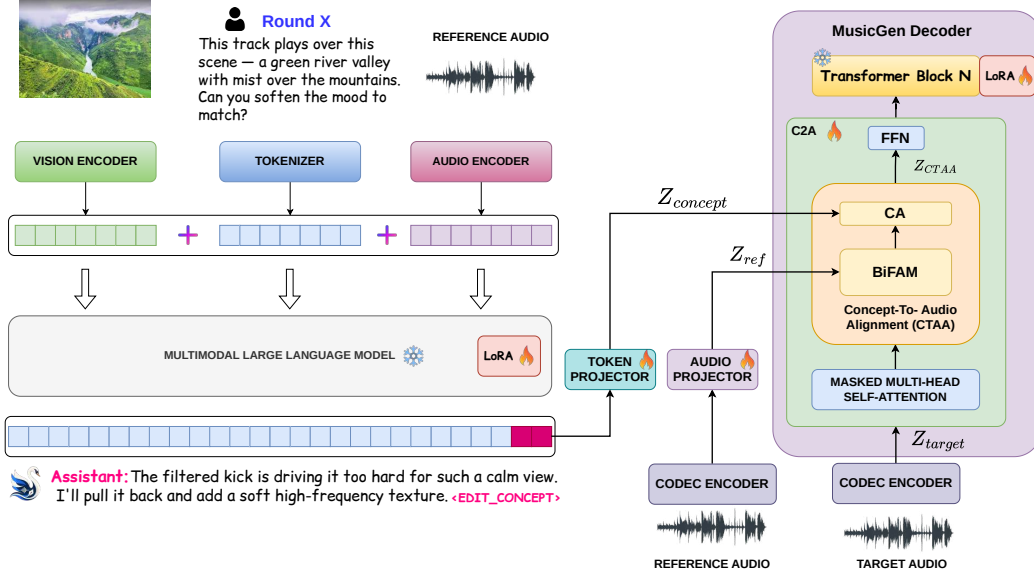


Fig. 1: Overall architecture of AURA. Given an image I , a multi-turn conversation T , and a reference audio X_{ref} , the multimodal LLM g_{mlm} produces a natural-language response together with $\langle \text{EDIT_CONCEPT} \rangle$ tokens that summarize the editing intent inferred from all modalities and the dialogue history. The projector g_{proj} maps the hidden states of these tokens into the music decoder’s conditioning space. The concept-to-audio module g_{c2a} fuses this concept embedding, together with the reference audio stream, into the hidden states of the frozen music decoder, which then generates the edited audio X_{target} .

modalities, while a MusicGen-based decoder applies the edit and preserves the unaffected content of the reference audio. As illustrated in Fig. 1, AURA comprises an MLLM, concept and audio projectors, a concept-to-audio (C2A) module plugged in the MusicGen decoder. The pretrained MLLM and MusicGen backbone parameters remain frozen, while their LoRA adapters, the projectors, and the C2A module are optimized during training.

2.2. Multimodal large language model

Prior music-editing models [12, 13, 11, 18] only use the current instruction and reference audio but do not allow dialogue input, limiting them to single-turn interaction. To support conversational editing, we employ the Thinker component of Qwen2.5-Omni [19] and introduce n_c instances of a special $\langle \text{EDIT_CONCEPT} \rangle$ token. The MLLM jointly processes the conversation history T , image I , and reference audio X_{ref} :

$$Y = g_{\text{mlm}}(I, T, X_{\text{ref}}), \quad (2)$$

where Y is a natural-language response optionally followed by the concept-token block. Because these tokens attend to all preceding modalities and dialogue turns, their final hidden states compactly encode the editing operation, target instrument, desired attributes, and conversational constraints. We denote these states by

$$\mathbf{H}_{\text{concept}} \in \mathbb{R}^{n_c \times d_{\text{llm}}}, \quad (3)$$

where $d_{\text{llm}} = 3584$. The concept states provide a compact semantic interface between multimodal understanding and audio generation. When no audio edit is required, the MLLM produces only a textual response and emits no concept tokens.

2.3. Concept and audio projectors

The concept and reference-audio representations originate from different feature spaces and must therefore be aligned with the MusicGen hidden space. For the concept tokens, we apply a LayerNorm followed by a two-layer MLP:

$$\mathbf{Z}_{\text{concept}} = \mathbf{W}_2 \text{GELU}(\mathbf{W}_1 \text{LN}(\mathbf{H}_{\text{concept}})), \quad (4)$$

where $\mathbf{Z}_{\text{concept}} \in \mathbb{R}^{n_c \times d_m}$ and d_m is the MusicGen hidden dimension. Layer normalization reduces sensitivity to changes in the scale of the MLLM hidden states during post-training. The resulting n_c vectors serve as the keys and values of the concept cross-attention in each decoder layer.

In parallel, the frozen EnCodec encoder extracts a frame-level representation of the reference audio. The audio projector maps this representation into the same hidden space:

$$\mathbf{Z}_{\text{ref}} = g_{\text{aud}}(E_{\text{codec}}(X_{\text{ref}})) \in \mathbb{R}^{L \times d_m}, \quad (5)$$

where L denotes the number of audio frames. The two projected representations have primary roles: $\mathbf{Z}_{\text{concept}}$ specifies what should be changed, whereas $\mathbf{Z}_{\text{ref}}$ provides frame-aligned information about the content that should be preserved.

2.4. Music decoder

The music decoder is based on MusicGen-medium [4], which autoregressively predicts four streams of EnCodec [20] codes at 50 Hz. Its pretrained backbone remains frozen to preserve the learned musical prior. Editing is introduced through trainable LoRA adapters and the proposed C2A module, which integrates the reference-audio representation $\mathbf{Z}_{\text{ref}}$ with the edit representation $\mathbf{Z}_{\text{concept}}$.

Frame-aligned reference fusion. The reference audio forms a generation-independent stream that is aligned with the autoregressive music stream. Let $\mathbf{H}_r^{(\ell)}$ and $\mathbf{H}_m^{(\ell)}$ denote the reference and music states, respectively, at decoder layer ℓ . The two streams use the same frozen attention projections:

$$(\mathbf{Q}_x^{(\ell)}, \mathbf{K}_x^{(\ell)}, \mathbf{V}_x^{(\ell)}) = \mathbf{H}_x^{(\ell)}(\mathbf{W}_Q^{(\ell)}, \mathbf{W}_K^{(\ell)}, \mathbf{W}_V^{(\ell)}), \quad x \in \{r, m\}, \quad (6)$$

Because the reference stream does not attend to the generated sequence, its representations can be computed once and cached throughout autoregressive decoding. Our **bidirectional feature-alignment module (BiFAM)** uses a shared, position-aligned query to retrieve information from both streams:

$$\begin{aligned} \mathbf{S}^{(\ell)} = & \alpha_r^{(\ell)} \text{Attn}(\mathbf{Q}_r^{(\ell)} + \mathbf{Q}_m^{(\ell)}, \mathbf{K}_r^{(\ell)}, \mathbf{V}_r^{(\ell)}; \mathcal{M}_{\text{full}}) \\ & + \alpha_m^{(\ell)} \text{Attn}(\mathbf{Q}_r^{(\ell)} + \mathbf{Q}_m^{(\ell)}, \mathbf{K}_m^{(\ell)}, \mathbf{V}_m^{(\ell)}; \mathcal{M}_{\text{causal}}), \end{aligned} \quad (7)$$

where $\mathcal{M}_{\text{full}}$ and $\mathcal{M}_{\text{causal}}$ denote full and causal attention masks; $\alpha_r^{(\ell)}$, and $\alpha_m^{(\ell)}$ are learned layer-wise weights.

Let $\mathbf{O}_m^{(\ell)}$ denote the decoder masked self-attention output. BiFAM modulates this output through a FiLM transformation:

$$\begin{aligned} \tilde{\mathbf{O}}_m^{(\ell)} = & \mathbf{O}_m^{(\ell)} \odot \left[\mathbf{1} + \tanh(g^{(\ell)}) \gamma^{(\ell)}(\mathbf{S}^{(\ell)}) \right] \\ & + \tanh(g^{(\ell)}) \beta^{(\ell)}(\mathbf{S}^{(\ell)}), \end{aligned} \quad (8)$$

where $g^{(\ell)}$ is a learned gate and $[\gamma^{(\ell)}; \beta^{(\ell)}]$ is produced by a rank-256 bottleneck MLP. The MLP output and gate are initialized to zero, making the transformation an identity mapping at initialization. This frame-aligned modulation preserves the reference structure without appending additional tokens to the autoregressive sequence.

Concept cross-attention. After reference fusion, the projected concept tokens are introduced as a shared semantic memory. At layer ℓ , the decoder attends to $\mathbf{Z}_{\text{concept}}$ as

$$\begin{aligned} \mathbf{C}_m^{(\ell)} = & \text{Attn}(\mathbf{Q}_c^{(\ell)}(\tilde{\mathbf{O}}_m^{(\ell)}), \mathbf{K}_c^{(\ell)}(\mathbf{Z}_{\text{concept}}), \mathbf{V}_c^{(\ell)}(\mathbf{Z}_{\text{concept}})), \\ \hat{\mathbf{O}}_m^{(\ell)} = & \tilde{\mathbf{O}}_m^{(\ell)} + \mathbf{C}_m^{(\ell)}. \end{aligned} \quad (9)$$

The same n_c concept vectors are shared across all decoder layers and generation steps. Rank-64 LoRA adapters are applied only to the key and value projections of the concept cross-attention. The resulting states $\hat{\mathbf{O}}_m^{(\ell)}$ are subsequently processed by the layer’s feed-forward network.

The zero-initialized BiFAM components and LoRA adapters preserve the behavior of the pretrained MusicGen model at initialization, allowing AURA to learn an editing residual rather than relearning audio generation. Overall, AURA contains 91M trainable parameters, while 1.9B backbone parameters remain frozen. At inference, classifier-free guidance performs an unconditional pass by replacing $\mathbf{Z}_{\text{concept}}$ with zero-valued memory while retaining $\mathbf{Z}_{\text{ref}}$. The resulting guidance direction therefore isolates the requested edit while holding the reference content fixed.

2.5. Objective function

AURA is trained end-to-end with a joint objective combining the language-modeling loss over assistant responses and the music decoder’s codebook cross-entropy:

$$\mathcal{L} = \mathcal{L}_{\text{MLLM}} + \mathcal{L}_{\text{music}}. \quad (10)$$

The language loss supervises multi-turn reasoning and the placement of $\langle \text{EDIT_CONCEPT} \rangle$ tokens, while the music loss shapes their representations through the projector. This joint supervision ensures that the learned tokens are both linguistically meaningful and useful for audio generation.

3. EXPERIMENTAL DETAILS

3.1. Experimental setup

Training dataset. We train on **66,539** conversational dialogues generated from the Slakh2100 training subset. Each dialogue contains a user turn (source audio + LLM-paraphrased instruction) with an assistant turn containing a typed edit-token block over 10 s EnCodec windows. Every sample is annotated with edit kind, a 10-class instrument label (94.5% coverage). For evaluation, we use 1000 in-domain samples from the Slakh test split (add/remove/extract at 32 kHz) and 1000 out-of-domain samples from MoisesDB with LLM-written instructions.

Implementation details. The MLLM emits $n_c = 9$ concept tokens, which condition the frozen MusicGen backbone with hidden size $d_m = 1536$. AdamW is used with a learning rate of 10^{-4} , an effective batch size of 16, and rank-64 LoRA ($\alpha = 128$) on the cross-attention key and value projections. Inference uses classifier-free guidance with a scale of 2.0.

Evaluation. We evaluate methods along four criteria: audio quality (FAD↓ [21], KL↓), instruction adherence (CLAP↑ [22]), edit correctness (P-Demucs↑ [23]), and content preservation (SSIM↑, SI-SDR↑, and SI-SDRi↑) [24]. SI-SDR and SI-SDRi are reported only for removal and extraction tasks, where deterministic targets are available. In our evaluation protocol, we conduct single-turn editing and multi-turn editing to assess the effectiveness of AURA.

3.2. Quantitative results

Single-turn editing. Table 1 shows that AURA consistently improves edit correctness and content preservation. On

Table 1: Comparison with SOTA music-editing methods, in-domain (Slakh2100-test) and out-of-domain (MoisesDB). SI-SDR is undefined for *add*. AURA substantially improves edit correctness and content preservation across both benchmarks.

Task	Model	Slakh2100-test (in-domain)							MoisesDB (out-of-domain)						
		FAD↓	CLAP↑	KL↓	SSIM↑	P-Dem.↑	SI-SDR↑	SDRi↑	FAD↓	CLAP↑	KL↓	SSIM↑	P-Dem.↑	SI-SDR↑	SDRi↑
Add	M2UGen	4.25	0.29	1.19	0.08	0.38	–	–	4.64	0.28	1.04	0.08	0.45	–	–
	Instruct-MG	2.01	0.28	0.65	0.37	0.48	–	–	3.84	0.18	0.84	0.32	0.51	–	–
	LeVo	2.30	0.28	0.82	0.11	0.35	–	–	3.92	0.25	0.87	0.10	0.44	–	–
	AURA	0.52	0.29	0.23	0.78	0.63	–	–	0.84	0.34	0.22	0.71	0.57	–	–
Remove	M2UGen	3.00	0.27	1.28	0.09	0.36	−46.01	−52.43	4.49	0.28	1.37	0.09	0.27	−43.21	−49.87
	Instruct-MG	1.42	0.37	0.50	0.43	0.55	−2.10	−8.53	3.55	0.20	0.71	0.37	0.49	−4.41	−11.03
	LeVo	2.20	0.35	0.98	0.10	0.31	−40.32	−46.74	3.50	0.31	1.01	0.10	0.26	−41.85	−48.47
	AURA	0.34	0.36	0.12	0.80	0.74	+11.32	+4.89	0.72	0.33	0.22	0.70	0.78	+9.16	+2.54
Extract	M2UGen	6.66	0.32	1.36	0.16	0.67	−45.47	−39.32	4.26	0.18	1.46	0.14	0.65	−43.61	−39.95
	Instruct-MG	5.32	0.44	0.74	0.27	0.79	−15.18	−9.05	5.20	0.18	0.72	0.28	0.79	−14.76	−10.39
	LeVo	5.95	0.29	1.76	0.12	0.44	−39.80	−33.67	4.44	0.17	1.41	0.09	0.40	−40.49	−36.11
	AURA	4.48	0.41	0.86	0.45	0.80	−7.62	−1.49	4.24	0.26	0.72	0.48	0.89	+2.67	+7.04

Table 2: Multi-turn editing benchmark MoisesDB (out-of-domain). Turns apply −drums, +bass, +guitar in sequence; each is scored against the exact intermediate target. The control (Ctrl) applies only the final edit to the original mix, without the intermediate turns.

Metric	Model	T1	T2	T3	Ctrl
SSIM↑	M2UGen	0.10	0.08	0.10	0.11
	Instruct-MG	0.42	0.32	0.22	0.32
	LeVo	0.08	0.07	0.08	0.08
	AURA	0.84	0.79	0.54	0.49
SI-SDR↑	M2UGen	−44.8	−45.9	−46.1	−47.1
	Instruct-MG	−0.9	−8.7	−19.3	−13.7
	LeVo	−38.7	−49.8	−52.7	−51.0
	AURA	+21.3	+12.3	+5.1	−5.2
SI-SDRi↑	M2UGen	−59.6	−48.3	−43.3	−44.4
	Instruct-MG	−15.7	−11.1	−16.5	−10.9
	LeVo	−47.3	−47.8	−46.6	−45.0
	AURA	+6.5	+9.9	+7.9	−2.4

Slakh2100, it more than doubles the best baseline SSIM for *add* (0.78 vs. 0.37) and uniquely achieves positive SI-SDR for *remove* (+11.32 dB). These gains generalize to MoisesDB, where AURA reduces FAD by 4–5× for *add* and *remove* and achieves positive SI-SDR for *extract*. Although its in-domain CLAP and extraction quality remain less consistent, AURA substantially improves preservation across tasks and domains. **Multi-turn editing.** Table 2 evaluates all methods under the same cascaded protocol. At each turn, M²UGen, LeVo, and Instruct-MG receive their own previous output as the new reference audio together with the current instruction, while AURA additionally receives the complete dialogue history. AURA maintains positive SI-SDR from +21.3 to +5.1 dB and positive SI-SDRi across all turns. In contrast, the strongest baseline, Instruct-MG, degrades from −0.9 to −19.3 dB, demonstrating greater error accumulation under sequential editing. Applying only the final instruction to the original mixture reduces AURA’s SI-SDR to −5.2 dB, indicating that the dia-

Table 3: Comparison of the effects of (EDIT_CONCEPT) tokens with fully LLM-generated token hidden states on the MoisesDB dataset. Best results are in **bold**.

Task	FAD↓		CLAP↑		SSIM↑	
	Text	Concept	Text	Concept	Text	Concept
Add	13.32	0.84	0.10	0.34	0.08	0.71
Remove	9.49	0.72	0.15	0.33	0.068	0.70
Extract	9.97	4.24	0.14	0.26	0.099	0.48

logue history is necessary to recover the intended editing state. Although AURA’s SSIM decreases from 0.84 to 0.54, it remains substantially more robust than the single-turn baselines.

3.3. Ablation study

Table 3 compares concept-token fusion with the decoder’s original full-text conditioning. Concept tokens consistently improve audio quality, suggesting that their compact semantic representations align more effectively with the music decoder and are less susceptible to irrelevant variation in the full-sentence hidden states. These results validate the importance of concept-token-based fusion.

4. CONCLUSION

We introduced AURA, a unified multimodal framework that brings conversational control to instruction-guided music editing. By routing all multimodal and dialogue context through a compact set of concept tokens, and by injecting the reference audio frame-aligned rather than as a style prompt, AURA separates *understanding what to change* from *producing the result* while keeping the pretrained music decoder frozen. Experiments on in-domain and out-of-domain benchmarks show large gains in content preservation and edit correctness over prior editors. The main limitation is stem extraction, where a generative decoder cannot match dedicated separation models; coupling AURA with an explicit separation front-end is a natural next step.

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